



Inspection-Certification

statutory inspection wind turbine generator

Report number	
Client	
Client contact	
Wind park	
Site address	
Type WTG	
WTG number	

Content

- 1.Certification
- 2.Reports/checklists
- 3.Foto Documentation

1.Certification

Certification report

The inspection/certification is made in accordance with manufacturer's instructions. The detailed inspection reports are listed in point 2.

Site	
WTG N°	
WTG type	
Year of construction	

Fundamental dates of stationary service cabin, ladder, fall arrest system, internal crane descender device and anchor points:

Designación/ Designation	Fabricante/ Manufacturer	Tipo/ Type	Número de serie/Serial number	Inspección pasó sin defectos/Inspection passed without defects	Inspección aprobada con pequeños defectos/Inspection passed with small defects	Inspección no aprobada/Inspection on not passed
Elevador/Service cabin						
Escalera torre/Ladder tower						
Linea de vida torre/Fall arrest system tower						
Polipasto/Internal crane						
Saca de rescatebc/Descender device						
Puntos de anclaje/Anchor points						

Summary

Inspectors	
Date	
Result of inspection	
Repair required	
Next inspection	

2.Reports/checklists

2.1 Inspection report service cabin

Content:

- 1 fundamental dates service cabin
- 2 checklist
- 3 notes

1 Fundamental dates service cabin:

Manufacturer service cabin	
Type	
Service cabin serial number	
Serialnumber Hoist	
Year build	
After 5 yrs or 200Hrs the 5-year Inspection required!	
Hour meter reading	
Inspection sticker is placed to follow up for next inspection?	

Summary:

Inspectors	
Date	
Result of inspection	
Repair required	
Next inspection	

Site, date

Responsible technician

2 Checklist:

1.BOTTOM - LANDING FLOOR CHECK

Service Lift	nk	ok	Refer notes, no.
Visual inspection of Service lift. Visual Damage?			
Connection Hoist and Stirrup is secured with nylock-nuts?			
All bolts and nylock nuts are fit and secured? (specific attention to Stirrup to SidePanels and bottom plate)			
Documents available in Service lift? (quick reference guide, operator manual, out of order sign,copy of yearly inspection report, electrical wiring diagrams, exploded views)			
All warning labels are still properly in place and readable.			
Gallery at bottom landing			
Galleries are correct? (see 38916-OM-E, section "Galleries According MD2006/42/EC")			
Locking system present on gallery?			
Visual inspect on damage and functioning			
All electrical cables strapped and not loose.			
MAIN SWITCH mounted on gallery			
LOCKING SYSTEM on gallery: transfer key can only be removed, if gate is locked.			
POWER ON LIGHT INSIDE SERVICE LIFT			
Under-"Bottom-Landing-Floor"-installations			
Installation double storage bin correct? (450mm - 500mm under bottom landing)			
Proper collectioning of electical supply cable in storage bin?			
Tension on the Guiding wire ropes is ok? (see position of red mark)			
Rubber protection in all holes?			
10kg Weights position and "Suspension wire rope" can rotate freely?			
Check function switches			
EMERGENCY STOP: up & down disabled; red safety integrity light is ON			
DOOR SWITCH: if door is open, up & down disabled; red safety integrity light is ON			
TRANSFER KEY SWITCH: if not switched on, then up & down disabled; red light is ON			
INSIDE steering: UP / DOWN control			
Control UP; safe zone (green light OFF); door locked			
BOTTOM LIMIT: down disabled, up still enabled			
BOTTOM LIMIT BY-PASS. (Pay attention to the weights on the suspension cable)			
ULTIMATE BOTTOM LIMIT (bott. Trip plate):up & down disabled;red safety integrity light ON			

NO POWER descent	nk	ok	Refer notes, no.
Check 'NO POWER' DESCENT' (Make sure speed will not increase)			
Overspeed check			
Manually trip overspeed and verify down direction is blocked (Electrically and with Brake)			
Check hand wheel for damage and check functionality to reset "overspeed reset knob"			
Overload check			
OVERLOAD FUNCTIONALITY: 240kg (nominal operation); 300kg (up & down disabled)			

2.1st FLOOR CHECK (1 person at 1st FLOOR)

Service Lift: 1 person at first level and 1 person inside the service lift.	nk	ok	Refer notes, no.
TOP LIMIT: up disabled, down still enabled			
ULTIMATE TOP LIMIT: up & down disabled; red safety integrity light is ON			
ELECTRIC SUPPLY PLUG: check for fixation and damage.			
Bottom floor connection bolts			
Actual Overspeed check (make sure gearbox is heated up)			
Disconnect motor-plug, (X6) Create overspeed (by opening brake), Down movement blocked, up still ok. Connect motor-plug and Reset			
Lubrication of suspension wire. (preventive action to avoid popping noise)			
Spray 1m of suspension wire and lift service lift up and down to let lubricated wire inside the hoist.			
OUTSIDE steering			
OUTSIDE steering: SET, then DOWN control (person on first level holds top limit to activate)			
OUTSIDE steering: SET, then UP control (stop with main switch)			

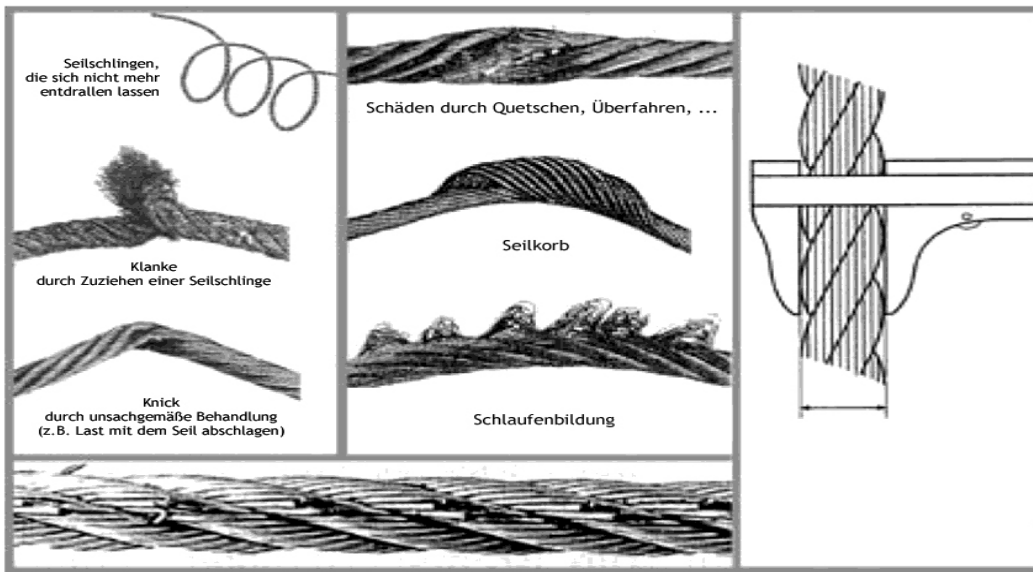
3.SPECIFIC VISUAL INSPECTION (during test-run)

Steel wire rope (dia 8,4mm), see 38913-OM-E, section "Steel Wire Rope"	nk	kok	Refer notes, no.
Check the condition of the 8.4mm Suspension wire rope.			
Check the condition of the 8.4mm Safety wire rope.			
Check the condition of both Guiding wires			
Interface with the surroundings			
Wire guides and wire fixes are working correctly? (easy passage; no shocks; no noise)			
Guiding wires do not show any damage.			
Verify that axial sway of service lift between two landings is acceptable and does not have any negative impact on the passage through the wire fixes on the landings: * Verify smooth passages in automatic material travel * Special attention should be given to inter distances above 20m: verify that tensioning of guiding wire ropes is set at approx. 400kg (= max. setting of tensioning system)			

Galleries on mid-level platforms	nk	kok	Refer notes, no.
Safezone (green light ON); door unlocked			
LOCKING SYSTEM on gallery: gate can only be unlocked with transfer key			
Galleries are correct? (see 38913-OM-E, section "Galleries According MD2006/42/EC")			

4.TOP PLATFORM CHECK

Gallery at top landing	nk	kok	Refer notes, no.
Safezone (green light ON); door unlocked			
LOCKING SYSTEM on gallery: gate can only be unlocked with transfer key			
Galleries are correct? (see 38913-OM-E, section "Galleries According MD2006/42/EC")			
Striker plate position			
Lift will stop on equal level as the gallery? If not change position of striker plate.			
Check striker plate fixation.			
Suspension beam check			
Connection of Guiding steel wires. (min. 2Ton Shackle and lock pins facing to the front)			
Connection of 8.4mm steel wires. (min. 2Ton Shackle and lock pins facing to the front)			
Condition of Shackles and check for wear.			
When replacing the Steel-wires check hole in Suspension beam for damage			



3 Notes:

2.2 Inspection report internal crane

Contents:

1 fundamental dates onboard crane

2 checklist

3 notes

1 Fundamental dates internal crane

Manufacturer onboard crane	
Type	
Year build	
Serial number	
Type plate and test badge	

Summary

Inspectors	
Date	
Result of inspection	
Repair required	
Next inspection	

2 Checklist

Generell	nk	ok	refer notes,N°
Reading of the operating hours			

Electrical equipment	nk	ok	refer notes,N°
Interrupt the voltage supply to the crane by removing the CEE plug from the top box in the powerhouse.			
Check that the circuit diagram is in position in the switch cabinet and is still legible			
Check that the warning signs on the switch cabinet, motor terminal box and power supply rails are in position and still legible			
Check all cables for signs of outer damage and firm attachment to the terminals			
Check all the screwed cable glands for signs of outer damage and a firm fit			
Open the switch cabinet cover and check the firm fit of the cables. Check whether the inside of the switch cabinet is free of dirt and/or humidity. Check that the snap-hook for hanging up the operating unit is in position and working.			
Check the operating unit for outer damage, a firm cable fit and make sure strain relief is working.			
Check the limit switches for signs of outer damage, a firm cable fit and ease of movement of the switching rockers			
Set up the voltage supply again. Check whether movement of the load crane corresponds to the symbols on the operating unit.			
Check that the motor brake is working. When the operating unit is actuated, the brake switches with an audible "click"			

Machine technology	nk	ok	refer notes,N°
Check the rope for mechanical damage such as breaks, pinching, bends or unwound strands by winding it off the drum completely and on again. Check the firm fit of the screws on the rope clamps on the drum. Grease the rope with a suitable rope grease with corrosion protection additives.			
Check the load hook for mechanical damage and check • the opening width of the hook opening • the perfect fit of the connection to the rope (rope eye stiffener and crimp sleeve) • the existence of rope weight and pressure spring • the existence and perfect condition of the tubular			
Carry out a visual inspection of the drive unit for damage and leaks. Listen for unusual noises during operation. Motor and gear are maintenance-free and do not require any lubrication.			
Use a feeler gauge to check the brake gap, it has to be 0.3 - 0.8 mm.			
Check the limit switch rockers to make sure they are present and move easily.			

Check that the crane bridge is safely locked in place and that the locking lever is a firm fit			
Check that the crab is safely locked in place and that the locking lever is a firm fit			
Check the deflection rollers and the respective components (rig pin, washer and spring cotter) to make sure they are in position and have no mechanical damage			
Check all the screw joints to make sure they are in position and a firm fit			
Check all the rollers on the bridge and on the crab for mechanical damage and to make sure they move easily by pushing the bridge and the crab by hand.			
Check all the stop buffers to make sure they are in position and have no mechanical damage.			
Check all the components to make sure the coating (paint, galvanizing) is in good condition. Suitable materials must be used to remedy damage to the corrosion protection. Bright steel parts e.g. roller bearings must be protected by a suitable preservation wax			

Ancor point safty equipment			
Load test			

3 Notes:

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2.3 Inspection report ladder,fall arrest system

Contents:

- 1 fundamental dates ladder,fall arrest system
- 2 checklist
- 3 notes

1 Fundamental dates ladder,fall arrest system

Manufacturer ladder	
Type	
Serial number	
Manufacturer fall arrest system	
Type	
Serial number	
Type plate and test badge	

Summary

Inspectors	
Date	
Result of inspection	
Repair required	
Next inspection	



Site ,date

Technician,responsible

Stamp,signature

2 Checklist:

Aluminum fixed ladders	nk	ok	refer notes,N°
Is ladder secure and tight to structure			
Are ladder brackets attached			
Are all fasteners in place			
Are all welds in good condition			
Are connections between rails ok			
Any bent rails/rungs			
Any cracked rails/rungs			
Is red rust/corrosion present			
Unobstructed climb path			
Is the overall condition safe			

Head anchor and cable termination	nk	ok	refer notes,N°
Is assembly secure and tight to Double-T			
Are attachment fasteners in place			
Is red rust/corrosion present			
Are swaged fittings in good condition			
Is eye bolt worn/cracked			
Are welds in good condition			
Any sign of wear or damage			
Is cable termination in good condition			
Does the overall condition appear safe			

Upright mount bracket assembly	nk	ok	refer notes,N°
Is assembly secure and tight to the structure			
Are channels secure to rungs			
Are all fasteners present and secure			
Is head anchor attachment secured			
Are welds free of cracks			
Are rungs bent or worn			
Is any red rust/corrosion present			
Does the overall condition appear safe			

Cable assembly	nk	ok	refer notes,N°
Does the cable show any flattened,frayed or kinked section/areas along the overall length			
Is the assembly secure to the head assembly			
Is the cable tight to the tower face			
Is the climb path free of obstuction			
Is the cable attached through the cable guides			
Does the cable show signs of red rust			
Does the cable show signs of electrical arcing/burns			
Does the overall condition appear safe			

Intermediate cable guides/stand Offs	nk	ok	refer notes,N°
Are cable guides provided every 20 to 40 ft.			
Are all fasteners present			
Are cable guides secure to the structure			
Are ladder rungs in good condition			
Are rubber grommets in good condition			
Is there any sign of red rust/corrosion			

Base anchor bracket assembly	nk	ok	refer notes,N°
Is assembly secure to the structure			
Are all clamp brackets/fasteners present			
Are the cable clips present and secure			
Is the cable taut at the assembly			
Is any rust/corrosion present			
Does the overall condition appear safe			

Cable safe climb pull test	nk	ok	refer notes,N°
Is the cable safe climb system secure			
Does the overall condition appear safe			

3 Notes:



2.4 Inspection anchor points

Contents:

1 fundamental dates anchor points

2 checklist

3 notes

1 Fundamental dates anchor points

Manufacturer anchor points	
Type	
Type plate and test badge	

Summary

Inspectors	
Date	
Result of inspection	
Repair required	
Next inspection	



Site ,date

Technician,responsible

Stamp,signature

2 Checklist:

Anchor point inspection	nk	ok	refer notes,N°
Inspect the anchorage device for legible serial number, stamping, deformation, corrosion, tear, or any kind of damage while it remains in its position			
Check that the anchorage device is tight in position			
Check that the bolt is going all way through the threaded hole of eye nut			
Inspect the steel surface for cracks and corrosion			
Record inspection			
List serial numbers. If anchorage device has no serial number, note the position in the turbine and no serial number			

Anchorage point	nk	ok	refer notes,N°
Anchorage point serial number Serial no/position: 77645/Middle section			
Anchorage point serial number Serial no/position: no serial number/Yaw deck			
Anchorage point serial number Serial no/position:			
Anchorage point serial number Serial no/position:			
Anchorage point serial number Serial no/position:			
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Anchorage point	nk	ok	refer notes,N°
Anchorage point serial number Serial no/position: 77645/Middle section			
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Anchorage point serial number Serial no/position :			

3 Notes:

2.5 Inspection descender device

Contents:

1 fundamental dates decender devices

2 checklist

3 notes

1 Fundamental dates descender devices

Manufacturer descender device	
Type	
Year build	
Serial number	
Type plate and test badge	

Summary

Inspectors	
Date	
Result of inspection	
Repair required	
Next inspection	



2 Checklist:

Descender device	nk	ok	refer notes,N°
Visual check of descender device			
Visual check rope			
Visual check sling and carabiner			
Do function test			
Fill document			
Seal back and put sticker on			

3 Notes:



2. Photo Documentation

Insert Images



General Service S.L.
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3. Additional Fields

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